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Performance-Influencing Factors and Causal Relationships of Construction Projects Using Smart Technology

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Abstract: With the advent of the Fourth Industrial Revolution, construction technology innovation through high-tech convergence is actively taking place, and the smart construction technology market is growing rapidly. However, as it focuses on the use of individual technologies, research into the factors that have a major impact on their intended effect is insufficient. Thus, this study investigates these factors and their mutual influences from various perspectives to promote the use of smart technology to improve construction projects. Ten performance-influencing factors were derived from four perspectives based on the balanced scorecard technique. In addition, based on a survey of projects that use smart technology, the current status of its application and performance characteristics were analyzed, and a causal relationship model among the factors was presented. This study provides a foundation for identifying major areas for the efficient use of smart technology and performance measurement, and it will contribute to the introduction and activation of smart construction technology.

Keywords: smart technology; project performance; influencing factors; causal relationship; balanced scorecard

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1. Introduction

Recently, the construction industry has been promoting productivity innovation and a paradigm shift toward knowledge and high-tech industries through the use of smart technology, which combines building information modeling (BIM), the Internet of things (IoT), big data, and artificial intelligence (AI), with traditional construction technologies [1]. According to MGI [2], the annual productivity growth rate in the construction industry over the past 20 years (1995–2015) has been only 1%, which is very low compared with the 3.6% average for the total manufacturing industry. The productivity gap is also rapidly widening, partly because the construction industry has the lowest level of digital innovation. If it were raised to the level in the manufacturing industry, productivity would be expected to increase by 25%. In addition, it would be expected to cause positive ripple effects such as cost and disaster reduction and increased added value. KPMG [3] investigated the application ratio of smart construction technology between the top (20%), central (60%), and bottom (20%) groups of construction and engineering companies and found a significant difference in the ratio of application among these groups, a gap that is expected to widen within the next 5–10 years. In other words, the use of smart technology based on digital innovation will have a significant impact on the competitiveness and survival of construction companies in the near future.

Against this background, major countries around the world are combining smart construction technology based on digital innovation to improve economic structures, create

jobs, and enhance industrial competitiveness. Representative government-led policies include Construction 2025 in the United Kingdom, I-Construction in Japan, and Construction 21 in Singapore. In addition, a specific roadmap to develop and apply smart construction technology according to construction life-cycle stages was presented in Korea in 2018 [1]. Meanwhile, in the United States, the private sector is pursuing ecosystem changes and innovation in the construction industry by revitalizing smart construction startups. As a result, attempts and related research are actively being conducted to incorporate smart technology into existing production processes with active participation from the government and the private sector.

A high-tech-based construction environment can serve as a driving force for dramatically improved construction performance, but to have the intended effect, organizations, processes, and infrastructure must be established. In other words, the effect is determined by the ability of the overall system to use technology in the field effectively rather than the by the type of technology applied. However, the most existing research related to smart construction technology focuses on applying and analyzing more efficient methods and frameworks [4–8]. In other words, research into factors that have a major impact on the intended effect of a specific technology is insufficient, and research into identifying and evaluating their relationship to project performance is also lacking.

Therefore, this study aims to investigate the major factors that affect the performance of construction projects using smart construction technology and analyze the causal relationship among these factors. The performance-influencing factors in this study, except for laws and policies, were limited to those that can be controlled and measured at the organizational level. These factors were constructed based on the balanced scorecard (BSC) methodology, which is widely used as a conceptual model for measuring corporate performance. The results of this study will contribute to providing a basis for the establishment of an organizational system and performance evaluation for the efficient use of smart construction technology.

2. Methods

The procedure for this study is shown in Figure 1. In the first step, we reviewed the use of smart technology in construction projects to demonstrate the need for this study. Next, we investigated studies on success factors and performance impacts after smart technology had been introduced. Finally, using the BSC technique, the perspectives and performance-influencing factors of construction projects using smart technology were constructed based on reviews and expert interviews.

In the second step, a four-part questionnaire survey, as shown in Table 1, was conducted to analyze the adequacy of the performance-influencing factors and their causal relationships. The questionnaire was distributed to the managers of engineering and construction companies through e-mail. In the survey, only managers with experience in carrying out smart technology-based projects were required to respond to the items in Parts 3 and 4.

First, the importance of each perspective and factor (Part 2) was surveyed, and the adequacy of the factors was investigated using reliability and factor analysis and the Student's *t*-test. Next, in Part 3, information on a representative project was investigated to provide an overview of the current use of smart technology and the level of application between top- and lower-ranked companies. Finally, satisfaction related to the influencing factors (Part 4) was measured using a 5-point Likert scale. Using statistical analysis, including the Student's *t*-test and regression analysis, we identified (1) deficiencies in the construction projects with smart technology, (2) satisfaction differences among the company groups and (3) causal relationships among the factors.

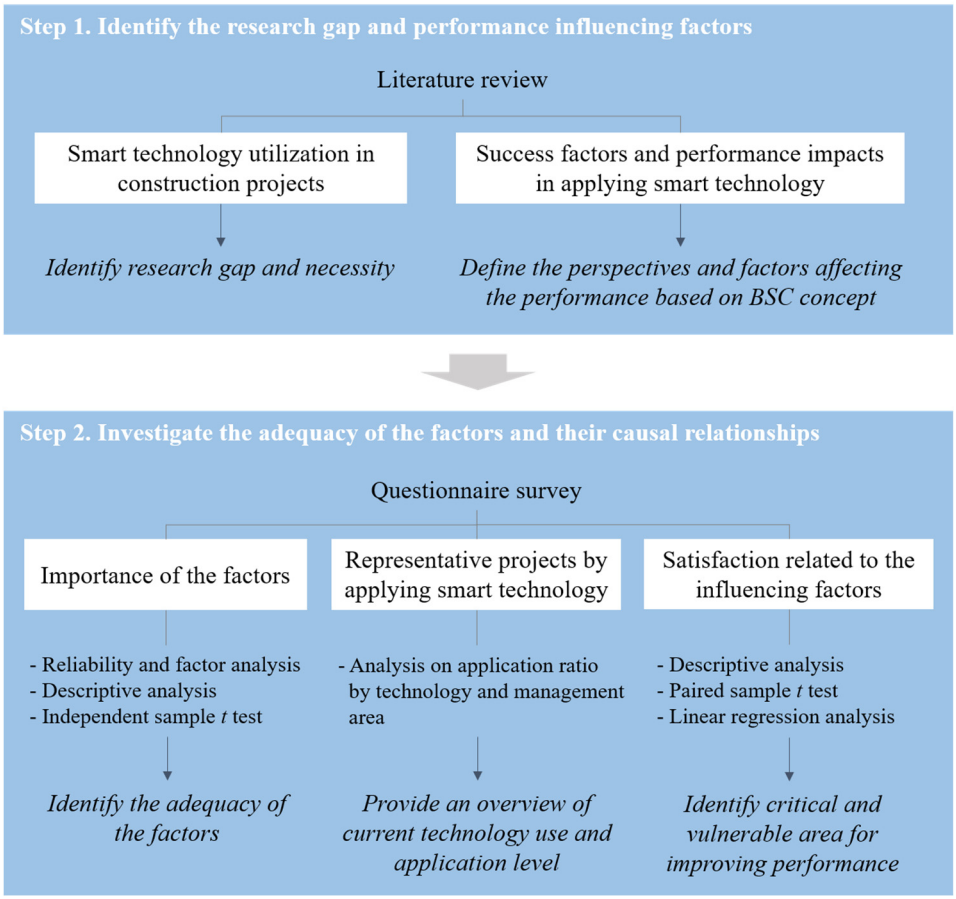


Figure 1. Research procedure.

Table 1. Questionnaire configuration.

Section	Contents	Remarks
Part 1	Basic information of respondents - company name - work experience in construction - experience in smart technology-based projects	
Part 2	Importance of perspectives and factors in Table 3	5-point Likert scale (from one indicating "not important at all" to five meaning "very important")
Part 3	Basic information on a representative project with smart technology - building use - construction cost - applied smart technology - management area using smart technology	
Part 4	Satisfaction related to the influencing factors in the representative project	5-point Likert scale (from one indicating "not satisfied at all" to five meaning "very satisfied")

3. Literature Review

3.1. Smart Technology in Construction Projects

Smart technologies are widely applied in all stages of construction production as shown in Table 2. They include (1) design automation and legal review using BIM and AI in the design stage [9–14], (2) real-time monitoring of construction resources using IoT

and drones and construction automation through intelligent construction equipment and robots in the construction stage [15–23], and (3) monitoring and operation optimization using sensing and big data analytics in the operation stage [24–29]. According to a scientometric review of smart construction sites in construction engineering and management by Liu et al. [30], the focus gradually shifted from traditional project performance-related concerns, such as hybrid information and performance evaluation, to practical applications of smart technologies in worker and construction site-associated areas.

Table 2. Research on smart technology use by construction stage.

Stage		Objective	Application Technology	References
Design	Design automation	Develop a generative BIM prototype for conceptual design automation	BIM, AI, genetic algorithm	[9]
		Propose a conceptual framework for the generative BIM platform	BIM, generative design	[10]
		Examine the applicability of the generative design concept in the building layout	BIM, visual programming	[11]
		Propose an integrated framework for the automated design generation of a modular house	BIM, coupled generative adversarial network (CoGAN)	[12]
	Automated code checking	Propose a fully automated semantic NLP-based automated compliance-checking system	BIM, semantic NLP algorithm, EXPRESS data processing algorithm, semantic-based logic reasoning algorithm	[13]
		Propose a semantic approach to automate the whole automated compliance-checking process	BIM, semantic NLP algorithm, ifcOWL, automatic SPARQL generation	[14]
Construction	Real-time monitoring	UAV system for autonomous mobile 3D mapping data acquisition in surveying earthworks project	Drone, laser scanning, autonomous vision-based infrastructure sensing	[15]
		Autonomous inspection of slope movement using drones	Drone, time series residual analysis	[16]
		Propose a real-time proactive warning system with equipment proximity sensing technology	Wireless radio frequency remote sensing, sensor actuation	[17]
		Develop a smartphone-based data collection system for concrete temperature monitoring	Smartphone, temperature sensor node, Zigbee	[18]
		Propose a mobile robotic platform for autonomous outdoor navigation	Robot, monocular SLAM, image segmentation technique	[19]
	Construction automation	Develop an autonomous location-aware mobile robot for brickwork	Robot, real-world sensor, mesh relaxation algorithm	[20]
		Propose 3D machine vision metrology for mobile construction robots	Robot manipulator, BIM, 3D scan, visual marker-based metrology,	[21]
		Propose an autonomous mobile robotic system for the surveillance of indoor environments	Robot, multi-sensor platform comprising a monocular camera, laser scanner, RFID	[22]
		Propose a framework of model-driven acquisition and analytics for automated construction progress monitoring	Drone, 4D BIM, appearance-based assessment	[23]
Operation	Monitoring	Propose a robot-aided tunnel inspection and maintenance system	Robot, lightweight robotic tool, vision, and laser telemeter	[24]
		Propose a cloud computing and big data processing architecture for real-time energy monitoring	Big data, cloud computing, smart meter, and environmental sensor	[25]
		Propose a building energy management control system with real-time adjustments	AI, IoT, reinforcement learning formulation	[26]
	Operation optimization	Introduce moving from BIM to digital twins for operation and maintenance	BIM, digital twin	[27]
		Propose a framework for implementing predictive building installation maintenance	IoT, machine learning, building automation system (BAS)	[28]
		Highlight the concept of big data analytics in predictive maintenance	Big data, vector autoregression, Granger causality	[29]

Despite ongoing practical efforts to use smart technology in construction engineering and management, organizational changes and efforts in various aspects are required. These can include addressing cost issues with information technology hardware and systems, optimizing resource allocation and task processes for immediate acquisition and information updating, improving collaboration among related engineers, and increasing user satisfaction and engagement [30]. In other words, it is necessary to make efforts in how to promote the performance of construction projects by utilizing the advantages of technology rather than the problem of applying any smart technology to the construction sites.

The results of the literature review showed that the research focus is still on which technology to apply (i.e., propose more efficient methods or show the effectiveness of case applications). By contrast, there is a lack of comprehensive surveys on issues that determine the performance of applications in the field. Accordingly, research is required to recognize factors that can promote the performance of smart technology-based projects. Therefore, this study is designed to identify factors and their mutual influences from various perspectives by performing construction engineering and management work at construction sites using smart technology. This can provide information on critical and vulnerable areas where project performance can be improved and contribute to a more effective organizational environment.

3.2. Performance Influencing Factors in Smart Technology-Based Projects

We conducted a literature review on success factors when introducing specific technologies such as RFID and BIM and evaluated the IT performance of construction companies. Based on BSC, Song et al. [31] determined 12 factors for measuring BIM project satisfaction from five perspectives: financial, customer, internal process, learning and growth, and informatization. From a regression analysis, it was found that a significant effect on improved satisfaction came from (1) profitability (the magnitude of financial effect compared with that of investment) and (2) organizational competency (professional manpower and collaboration). Shin et al. [32] developed an IT BSC-based evaluation system for BIM performance measurement and derived 13 critical success factors from four perspectives. As a result of a causal relationship analysis between the perspective and key success factors, it was found that the operational efficiency perspective and the BIM-based work process factor within that perspective had the greatest influence. Seo [33] analyzed the success factors of RFID system-built companies and their impact on performance and found that organizational factors such as CEO support, the cooperation system, member participation and understanding had a significant impact on performance as did technical factors such as system construction and compatibility. Cha and Yu [34] presented indicators for measuring corporate performance according to construction informatization and highlighted profitability, organizational capacity, and market share as major criteria. Li and Wang [35] derived 20 measures for evaluating IT benefits in construction companies and found information system capabilities, customer satisfaction, employee capabilities, and productivity to be relatively important.

The BSC technique, which was first presented in 1992 by Kaplan and Norton [36] and widely used as a conceptual model for performance measurement, was used in most previous studies. BSC evaluates a company's performance from a financial perspective representing execution results, customer satisfaction, internal business procedure, learning and growth. It follows the logic that the innovation and growth of employees and organizations lead to improved processes that increase customer satisfaction and consequently improves financial performance [37]. Thus, the four perspectives have a causal relationship, and performance can be evaluated while maintaining a balance.

Based on the literature review and the BSC concept, this study consists of four perspectives: (1) a business contribution representing the overall project contribution against the investment and use of smart technology, (2) a user orientation that evaluates the satisfaction of clients and companies, (3) an operational excellence that evaluates the efficiency of smart technology in the field, and (4) a growth and innovation that evaluates the level of an

organization's capacity and preparation. In addition, 13 factors corresponding to the four perspectives were derived, and then 10 factors were finally derived by integrating and adding some factors through expert interviews (Table 3).

Table 3. Factors influencing the project performance using smart technology by perspective.

Perspectives	Factors	Definitions
(A) Business contribution	(A1) Profitability	Cost effectiveness on investment
	(A2) Productivity	Schedule compliance and duration reduction
	(A3) Management performance	Improvement in targeted management performance
(B) User orientation	(B1) Satisfaction	Satisfaction of owner and organization members
	(B2) Marketability	Project contract and marketing effectiveness
(C) Operational excellence	(C1) Investment	Investment and support for related systems
	(C2) Work efficiency	Decision structure and procedure deployment
(D) Growth and innovation	(D1) Organizational capability	Professional manpower retention and cooperation system
	(D2) Technology utilization ability	Technology understanding and proficiency
	(D3) Education	Adequacy of education and training program

4. Results

A questionnaire survey was conducted to identify critical and vulnerable areas for using smart technology to enhance project performance. For Parts 1 and 2, 166 questionnaires from a total of nine companies were analyzed, excluding those judged to be missing or unreliable (collective answers). The average experience in the construction industry was about 19.2 years, and 50% of the respondents had used smart construction technology. Thus, a total of 83 questionnaires were used for the analysis of Parts 3 and 4.

4.1. Importance of Performance Influencing Factors

The consistency of survey responses was measured through Cronbach's α, and the criterion for response reliability was judged to be 0.6 or higher [38]. The results showed that business contribution (0.661), operational excellence (0.652) and growth and innovation (0.797) were higher than baseline, while reliability was low at 0.476 from the user orientation perspective. Accordingly, marketability had the lowest importance among all factors (see Table 4) and was not grouped because the exploratory factor analysis was removed. However, when the item was deleted, the Cronbach's α value increased to 0.690 for improved management performance. The difference was not large, and the importance of the factor was the largest, so it was judged not desirable to exclude it.

Table 4. Importance of perspectives and factors.

Category	ID	Mean	Standard Deviation	Skewness	Kurtosis	Rank
Perspectives	A	4.08	0.778	−0.450	–0.383	1
	B	3.86	0.901	−0.377	−0.412	4
	C	3.90	0.912	−0.571	−0.173	2
	D	3.89	0.975	−0.602	−0.237	3
Factors	A1	3.85	0.878	−0.298	−0.416	7
	A2	3.94	0.872	−0.382	−0.647	3
	A3	4.03	0.774	−0.290	−0.654	1
	B1	3.99	0.846	−0.342	−0.741	2
	B2	3.71	0.902	−0.047	−0.877	10
	C1	3.75	0.899	0.016	−0.996	9
	C2	3.89	0.799	−0.165	−0.672	5
	D1	3.87	0.891	−0.204	−0.908	6
	D2	3.92	0.888	−0.319	−0.799	4
	D3	3.80	0.916	−0.312	−0.727	8

Table 4 shows the results of a descriptive statistical analysis according to the importance of each perspective and factor. Because the absolute values of both skewness and kurtosis were less than 1, there was no problem determining the average relationship among the variables as they did not deviate significantly from normal. From this perspective, the importance of business contribution (A) was the greatest, followed by operational excellence (C). Regarding the factors, management performance (A3) had the highest importance, meaning that its targeted improvement from applying smart technology was considered a key factor in project performance. By contrast, the importance of user orientation (B) was the lowest, but the satisfaction (B1) factor was the second highest, so the improved satisfaction of the owner and organization members from the smart technology had a notable impact on performance. The importance of work efficiency (C2) and technology use ability (D2) was the highest for operational excellence and growth, and innovation, respectively.

In addition, the average difference among managers with experience in carrying out construction projects with smart technology (n = 83) and managers without such experience was analyzed using an independent sample Student's t-test. There was a relatively large average difference in factors such as organizational capability and investment, but no statistically significant difference was found between the groups, so the importance of perspectives and factors was considered similar regardless of experience. Consequently, all the factors presented in Table 2 showed high importance with an average of 3.5 or more, and the factor classification by perspective, excluding the marketability (B2) factor, was properly conducted. Thus, they are critical factors that affect the performance of smart technology-based projects.

4.2. Analysis of Performance Characteristics and Causal Relationships

4.2.1. Project Characteristics

Housing (55.1%) and office buildings (29.2%) accounted for the majority of building use, and the average construction cost was about USD 211 million (USD 7.5 million to 1.1 billion). The smart technology was largely classified into data collection, visualization, and data processing and information sharing based on the research by Kim et al. [39]. The application of detailed technology applied according to each classification is shown in Figure 2. The application of data collection was the most prevalent, while the use of processing and sharing using the collected data was relatively limited. As for the detailed technologies, the most actively applied were BIM, drones, intelligent CCTV and 360 cameras, while big data, augmented/virtual reality, and AI had limited implementation.

In addition, the management category of smart construction technology use was, in order, safety (30.9%), schedule (28.7%), quality (22.7%), cost (9.9%), and material and outsourcing management (5.5%). As a result, most of them actively used it in safety and schedule management tasks. In addition, each task had a highly specific relation: schedule management with drone and BIM, safety management with CCTV/camera and drone, and cost management with BIM.

As shown in Table 5, the application ratio of smart technology by management area was analyzed by dividing the respondent companies into two groups: (1) the top 100 companies in the local construction capacity ranking (n = 60) and (2) companies ranking below the top-100 (n = 23). In this study, the ranking was based on data released by the Korean Ministry of Land, Infrastructure and Transport in 2022 by evaluating construction capabilities based on performance, management status, technical capabilities and company credibility [40]. Except for the intelligent CCTV/360 camera, the technology application ratio of the top companies was significantly higher than that of the lower-ranked companies; in particular, there was a notable difference in the application ratio of 3D laser scanning and BIM. In addition, in the top companies, the use of data processing and sharing technologies using big data and AI was more prevalent. In management tasks, most of the lower-ranked companies were biased toward the use of technology for safety management, while the top-ranked companies used it evenly in the areas of schedule, safety, and quality manage-

ment. As a result, lower-ranked companies continued to use data collection technology such as CCTV and drones to perform safety management tasks, while the top-ranked companies used smart technology—-from data collection to visualization and information sharing—-in a wider range of management areas.

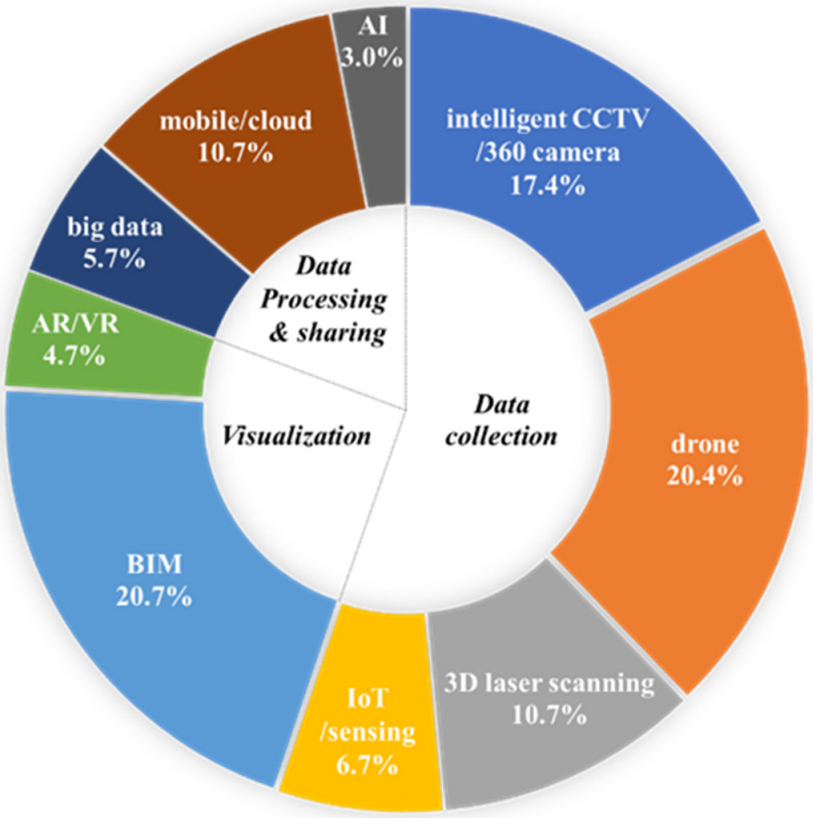


Figure 2. Application ratio of smart technology.

Table 5. Application ratio of smart technology and management area by group.

Category		Application Ratio by Group (%)		Remark (A–B)
		Top-Ranked Companies (A)	Lower-Ranked Companies (B)	
Technology	Intelligent CCTV/360 camera	58.3	70.8	−12.5
	Drone	76.7	62.5	14.2
	3D laser scanning	48.3	12.5	35.8
	IoT/sensing	26.7	16.7	10.0
	BIM	80.0	58.3	21.7
	AR/VR	20.0	8.3	11.7
	Big data	23.3	12.5	10.8
	Mobile/cloud	41.7	29.2	12.5
	AI	13.3	4.2	9.2
Management	Schedule	68.3	45.8	22.5
	Safety	65.0	70.8	–5.8
	Quality	53.3	37.5	15.8
	Cost	26.7	8.3	18.3
	Material and outsourcing	13.3	8.3	5.0

4.2.2. Performance Characteristics

Table 6 shows the survey items used to investigate satisfaction with the performance influencing factors. Based on the results in Section 4.1, an item related to the marketability factor was excluded from the analysis.

Table 6. Survey items for investigating satisfaction with the performance influencing factors.

Factors	Items
(A1) Profitability	The use of smart technology in project execution had a financial effect more than the investment amount.
(A2) Productivity	The use of smart technology contributed to project schedule compliance and duration reduction.
(A3) Management performance	It contributed to the improvement of the intended management performance through the application of smart technology.
(B1) Owner satisfaction	The owner was satisfied with the results of applying smart technology.
(B2) Satisfaction of organization members	Organization members were satisfied with the performance of construction management tasks using smart technology.
(C1) Investment	Investment in the development and deployment of smart technology-related systems needed to carry out the project was appropriate.
(C2) Work efficiency	When performing management tasks using smart technology, the decision-making structure was clear, and procedures were properly in place.
(D1) Organizational ability	It appropriately had the professional manpower necessary to perform management tasks using smart technology.
	In using smart technology, a cooperative system with the information system and related departments was well established.
(D2) Technology use ability	It had the ability to perform management tasks using smart technology on its own.
(D3) Education	Education and training for the use of smart technology were properly conducted for practitioners.

As shown in Table 7, the owner satisfaction (B1, 3.65) factor in the user orientation perspectives was the highest. Similarly, satisfaction from organization members (B2, 3.61) management performance (A3, 3.63) and profitability (A1, 3.42) from a business contribution perspective were also high, while satisfaction with factors (D1–D3) from a growth and innovation perspective was relatively low. In other words, construction managers believed that the positive effects of using smart construction technology went beyond investment to the improvement of management performance and the satisfaction of owners and organization members. On the other hand, results for organizational capabilities and education support were relatively insufficient.

Table 7. Satisfaction by factor on the project using smart technology.

Factors	Mean	Standard Deviation	Skewness	Kurtosis	Rank
A1	3.42	0.960	−0.344	−0.333	4
A2	3.12	1.034	−0.378	−0.096	7
A3	3.63	0.915	−0.254	−0.247	2
B1	3.65	0.938	0.031	−0.960	1
B2	3.61	1.030	−0.634	−0.058	3
C1	3.31	1.029	−0.183	−0.289	5
C2	3.13	1.106	−0.211	−0.442	6
D1	2.98	0.970	0.153	−0.0301	9
D2	3.04	1.113	−0.125	−0.514	8
D3	2.87	0.929	−0.011	−0.093	10

From the results of Tables 4 and 7, the difference between importance and satisfaction for each factor was analyzed to identify deficiencies in the construction projects with smart technology. As a result of the paired sample Student's *t*-test, the average satisfaction

comparison with importance was low for all factors, and there was a statistically significant difference. In particular, there was a relatively large difference in the factors for growth and innovation and operational excellence. Accordingly, the overall organizational system and competency level for the effective use of smart technology were found to be insufficient, and the achievement of performance such as productivity improvement was also considered somewhat poor.

The average difference in performance was verified between the top- and lower-ranked companies. Table 8 shows the results of an independent sample Student's *t*-test on the performance from the BSC perspective. As shown in Table 8, there was a statistically significant difference ($p < 0.05$) only in the growth and innovation perspective. Accordingly, the top-ranked companies were ahead in investment and efforts for growth and innovation. In addition, as a result of verifying the average difference in performance by factor, there was a statistically significant difference ($p < 0.05$) in organizational capability (D1) and technology use ability (D2) in the growth and innovation perspective, indicating that the difference between the groups was caused by these factors.

Table 8. Comparison of performance differences between top- and low-ranked companies.

Perspectives	Group [1]	Mean	Standard Deviation	*t* Value	*p*
Business contribution	A	3.1667	0.79855	−1.536	0.128
	B	3.4778	0.85341		
User orientation	A	3.5625	1.03538	−0.448	0.656
	B	3.6583	0.82077		
Operational excellence	A	2.9792	1.12751	−1.454	0.150
	B	3.3167	0.88761		
Growth and innovation	A	2.6042	0.87823	−2.465	0.016
	B	3.1125	0.84425		

[1] A = lower-ranked companies, B = top-ranked companies.

4.2.3. Causal Relationship

Based on the satisfaction results of the representative projects, we analyzed the impact of other perspectives on business contribution (A) through a linear regression analysis. It had a significant effect on improving the performance of the business contribution perspective through the use of smart technology in the growth and innovation (D) and user orientation (B) perspectives (Table 9). In other words, the proper establishment of organizational capabilities and education, rather than the appropriateness of investment and the operational procedure, had a somewhat significant impact on improved performance. Accordingly, efforts should be made to improve the level of factors from the growth and innovation perspective, which are currently somewhat insufficient.

Table 9. Result of regression analysis between perspectives.

Variables	Unstandardized Coefficients		Standardized Coefficients	*t*	*p*	F	Adjusted R^2
	B	Standard Error	B				
(B) User orientation	0.357	0.093	0.372	3.850	<0.001	26.262 ($p < 0.001$)	0.496
(C) Operational excellence	−0.011	0.105	−0.013	−0.109	0.913		
(D) Growth and innovation	0.433	0.111	0.451	3.896	<0.001		

We also analyzed the relationships between factors based on the conceptual causality of BSC. For example, two multiple regression models (1 and 2 in Figure 3) were created using three factors within the growth and innovation perspective as independent variables and two factors within the operational excellence perspective as dependent variables. Accordingly, a total of 15 models were derived, as shown in Figure 3.

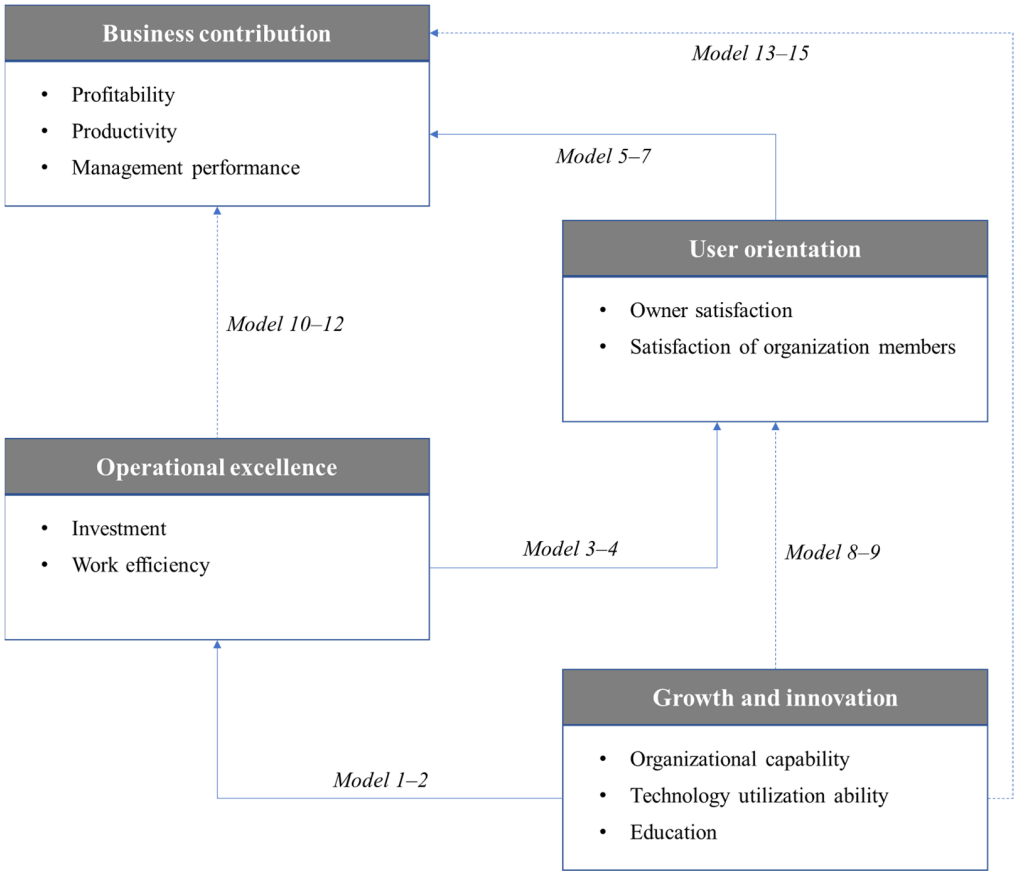


Figure 3. Regression models based on conceptual causality of BSC.

Figure 4 shows the causality model between factors from a regression analysis, where causality was based on statistically significant variables in each model. The explanatory power of each was somewhat low (6.9–50.4%), but all regression models were statistically significant ($p < 0.05$). However, no variables had a significant effect in the models 8, 10, and 13. First, from the perspective of growth and innovation, organizational capability (D1) acted as a major factor that affected both factors (i.e., investment (C1) and work efficiency (C2)) from an operational excellence perspective. That is, the better the professional manpower and cooperation system, the more appropriate the work procedure and investments. Appropriate education (D3) also affected efficient work performance (C2). Factors C1 and C2 from the operational excellence perspective were relatively closely related to the improvement of member satisfaction (B2). In addition, the higher the member satisfaction (B2), the better the overall business contribution (i.e., profitability (A1), productivity (A2), and management performance (A3)). Concerning the impact of business contribution, productivity (A2) was mainly affected by organizational capability (D1), investment (C1), and member satisfaction (B2), while the factors of technology use (D2), investment (C1), and satisfaction (B1 and B2) had a major impact on management performance (A3). Consequently, establishing appropriate professionals, cooperative systems, and education promotes systematic business procedures and system investment, increasing the productivity and performance of the management work.

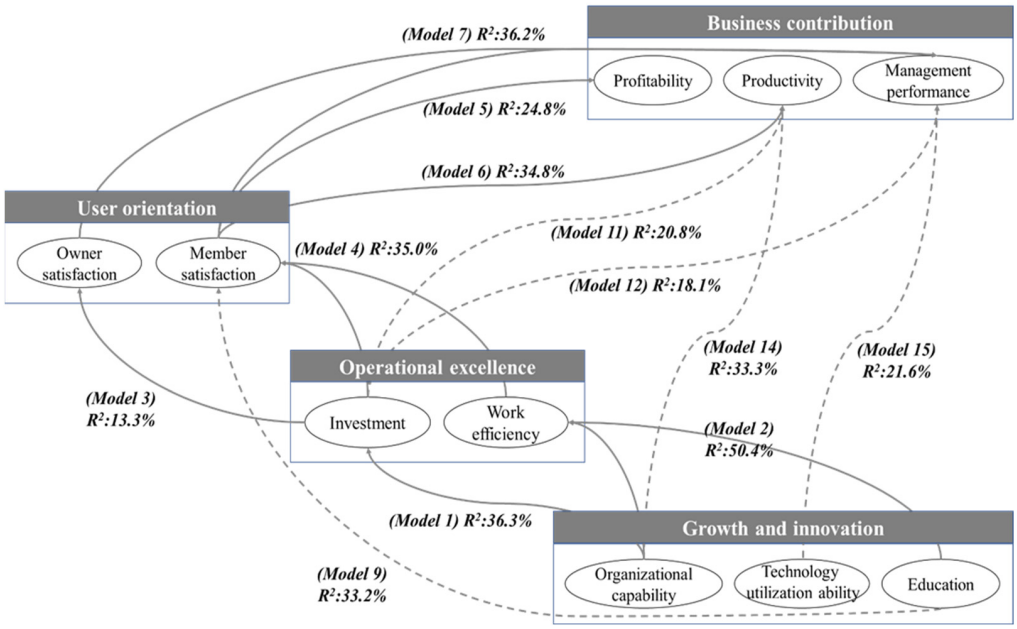


Figure 4. Causality model between factors by regression analysis.

5. Discussion and Conclusions

In recent construction projects, efforts have been made to solve productivity stagnation, safety, and quality problems caused by a lack of skilled workers through the use of smart technologies. However, their application requires organizational changes in various aspects, which greatly influences how well they are used. Thus, this study investigated major factors that affect the performance of construction projects with smart technology and their causal relationships.

Based on the BSC methodology, 10 factors were derived from four perspectives. Except for one, the remaining factors were not only properly classified by perspective but also identified as major factors affecting performance. Next, the performance status and causal relationships among factors were presented based on the survey of the actual projects. Compared with lower-ranked companies, the top-ranked showed a relatively high application rate of smart technology over a wider range of management areas. This result supported the KPMG survey [2] and showed that using smart technology had an important impact on competitiveness. From a comparison of the results of importance and satisfaction with performance influencing factors, the level of preparation for the effective use of smart technology in Korea was found to be insufficient. This supports the current focus on which technology to apply. As a result of the performance impacts and causal relationships, proper organizational capabilities and educational support for the use of smart technology in the field play a key role in improving performance. Song et al. [31] found that organizational capabilities significantly improved BIM project satisfaction, and Yu et al. [41] confirmed that a learning and growth perspective centered on the investment in technical personnel was central to the performance management of construction companies. Thus, this result agreed to some extent with the findings of previous studies and showed that investment in those areas can provide a critical competitive advantage. In addition, similar to the results presented by Yu et al. [41], the top-ranked companies were found to be ahead in investing in these factors for growth and innovation compared with the lower-ranked companies. Therefore, more active efforts to improve the capabilities including securing professional manpower, establishing a cooperative system, and improving educational support will contribute to improving the overall project performance and securing corporate competitiveness.

The results of this study will be used to establish a basis for an organizational system for the efficient use of smart technology and performance evaluation. However, the

performance characteristics and causal model presented in this study have limitations in that they were based on the practitioner's qualitative evaluation of just one country. Thus, although the proposed performance influencing factors can be applied regardless of country, additional comparative studies on performance characteristics need to be conducted to provide data to understand the difference in the national levels of preparation for the effective use of smart technology. Performance indicators must also be developed that can be quantitatively measured for each factor. In addition, continued research is needed on the development of a performance evaluation system through weighting to reflect the impact of each factor on performance.

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